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March 29, 2006

Via Certified Mail
Return Receipt Requested
7004 2890 0003 2049 5856

Subject: Cemex, Inc., Lyons Quarry, Permit No. M-1977-208, Technical Revision No. 7 (TR-07), pH and Selenium Treatment of C-pit Water

TR-07 ✓

Dear Erica:

This letter and attachment, along with a check for \$188, comprises Technical Revision No. 7 (TR-07) for the above referenced permit. TR-07 was required by TR-06 and addresses the contingency of treating C-pit water for selenium and pH.

If you have any questions regarding this submittal, please feel free to contact me at (303) 823-2115.

Sincerely,

Eric S. Hodek
Environmental Manager

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Division of Minerals and Geology

Encl

Cc: Paul Banks, Banks and Gesso, LLC
Al Notary, Brown and Caldwell

**TECHNICAL REVISION No. 07
SELENIUM REMOVAL SYSTEM
CEMEX LYONS PLANT
C-PIT WATER**

As TR-07 to the permit for the CEMEX Lyons Plant in Lyons, Colorado, the Division of Minerals and Geology (DMG) requires CEMEX to prepare a design and cost estimate to support the calculation of a bond to cover the cost of implementing a treatment remedy for water remaining in C-Pit in the unlikely event that CEMEX defaults and ceases operations.

This document presents a conceptual design and cost estimate for a biological water treatment system to remove selenium from water that has accumulated in C-Pit at the CEMEX Lyons Plant. Continuous-flow, bench-scale treatability testing has demonstrated that the process can reduce C-Pit water selenium concentrations from 430 micrograms per liter ($\mu\text{g/L}$) to below 60 $\mu\text{g/L}$. The average selenium removal rate exceeded 90 percent.

DESIGN OBJECTIVES

The proposed system will meet the following objectives:

- Remove as much residual water as practicable from C-Pit
- Leave no final free water surface in C-Pit
- Reduce the selenium concentration of the water to 60 micrograms per liter ($\mu\text{g/L}$) or lower
- Be operable independent of the existing cement manufacturing operations.

DESCRIPTION OF PROCESS

Figure 1 shows the conceptual design of the biological treatment process. Water is conveyed by the C-Pit pump via the C-Pit line to the treatment area. Concentrated sulfuric acid is injected by a metering pump into the C-Pit line upstream of an in-line mixer, thereby reducing the influent pH to the neutral range of 6-9. The C-Pit pump, C-Pit line, acid storage tank, acid metering pump and in-line mixer are all existing items that are in the process of being installed under the terms of Technical Revision TR-06, dated December 2, 2005.

The bulk of the treatment process is provided by Zenon Environmental (Zenon) of Oakville, Ontario, Canada (<http://www.zenon.com>). Proprietary nutrient solution provided by Zenon is injected by a metering pump into the C-Pit line. This mixture flows to the bottom of Reactor 1, which is a rented 21,000-gallon tank filled with activated carbon and inoculated with selenium-reducing bacteria. Effluent is drawn from the top of this tank and conveyed by gravity to the bottom of an identical Reactor 2. Additional nutrient is pumped into the Reactor 2 feed.

Effluent from Reactor 2 flows by gravity into a small effluent tank for sampling and discharge. Treated water is pumped from this tank for use at the plant for dust suppression during plant decommissioning and final reclamation of the plant site and any unreclaimed mining sites

DESIGN FLOW RATE

The system is designed to treat up to 100 gallons per minute (gpm) of C-Pit water. Since the process is biological, the treatment rate is expected to be temperature-dependent, so the pumping rate will need to be reduced during colder months. The system as designed will treat an average of 67 gpm, or 35 million gallons per year.

OPERATIONS

After construction of the system, Zenon will inoculate the reactors with bacteria. The pumps will then be started, and acidified C-Pit water with nutrients will be fed to the reactors. Reactor 2 effluent should be returned to the C-Pit until analytical testing shows that selenium is being removed sufficiently for discharge. Once testing shows that the system is stable and the treatment process is effective, processed water can be used for dust suppression and site reclamation activities.

Normal operation will consist of maintaining sufficient acid and nutrient supplies and checking the operation of process pumps. Effluent samples should be collected and analyzed periodically for selenium to monitor water quality.

With pumping, treatment, and discharge, the water volume and surface area in C-Pit will decrease to some small residual amount, beyond which it will become impossible to continue pumping. At that point, the treatment system will be removed from the site, and sufficient porous fill material will be placed in C-Pit to eliminate a free water surface. For the purpose of cost estimating, the volume of imported fill material has been assumed at 2,000 cubic yards (5 feet deep over one-quarter acre). The placement of this fill is the final element in the treatment process required by TR-07. Final reclamation of C-Pit and the plant site are covered by other requirements and bonding under CEMEX's permit with DMG.

LIMITATION

A continuous-flow, bench-scale test system operated at Zenon's laboratory for 14 weeks produced an effluent averaging 33 $\mu\text{g/L}$ selenium. The maximum effluent concentration was 54 $\mu\text{g/L}$. These statistics do not include one outlier that was reported at 576 $\mu\text{g/L}$ (higher than the system feed concentration) or data from the first month of operation while the system was acclimating to the water.

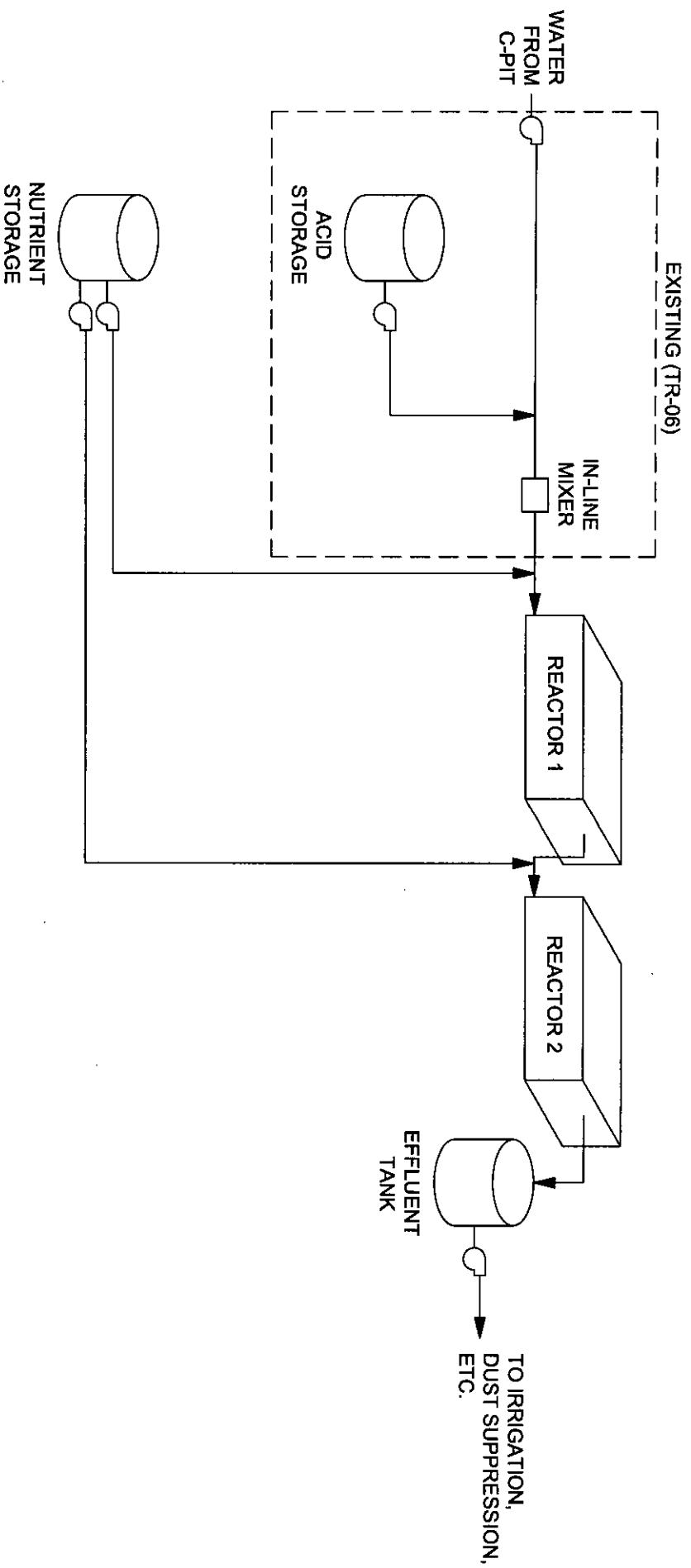
The system did not achieve the agricultural standard of 20 $\mu\text{g/L}$ selenium. However, the residual selenium in the effluent is apparently not biodegradable, since testing showed that doubling the residence time of biological treatment did not remove the residual.

COST ESTIMATE

Zenon estimates that a system sized for a nominal 100 gpm flow rate has a total cost of \$600,000 (capital cost plus operation for approximately one year to treat 35 million gallons). This amount includes pumps, reactor vessels, activated carbon, inoculum, acid and nutrient delivery systems, instrumentation, controls, electrical equipment, a building, installation, engineering support and supplies. CEMEX is presently purchasing and installing the acid tank and neutralization system under TR-06, so this portion of the process could be subtracted from the \$600,000 figure, but its inclusion will provide a contingency for the project.

Labor for operating the system is estimated at under \$20,000 (two hours per day at \$25 per hour for one year).

Placing 2000 cubic yards of fill in the residual C-Pit pool at the end of the treatment period will cost an additional \$5000. Therefore, the total cost to eliminate a free water surface from C-Pit is estimated at \$625,000.



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FIGURE 1 - CONCEPTUAL DESIGN FOR C-PIT WATER
BIOLOGICAL SELENIUM REMOVAL SYSTEM
CEMEX - LYONS, COLORADO